

PARAMETERS NEEDED FOR THE 2-WIRE EXTERNAL CONTROL MECHANISM

Parameter	Explanation	Settings	Factory setting
Pr.00	Source of Frequency Command	00: Master frequency determined by digital keypad 01: Master frequency determined by 0 to +10V input on AVI terminal with jumpers 02: Master frequency determined by 4 to 20mA input on ACI terminal with jumpers 03: Master frequency determined by RS-485 communication port 04: Master frequency determined by potentiometer on digital keypad	00
Pr.01	Source of Operation Command	00: Operation determined by digital keypad 01: Operation determined by external control terminals, keypad STOP is effective 02: Operations determined by external control terminals, keypad STOP is ineffective 03: Operation determined by RS-485 communication port, keypad STOP is effective 04: Operation determined by RS-485 communication port, keypad STOP is ineffective	00
Pr.38	Multi-function Input Terminal (M0,M1)	00: M0: FWD/REV, M1: REV/STOP 01: M0: RUN/STOP, M1: REV/FWD 02: M0, M1, M2: 3 wire operation control mode	00

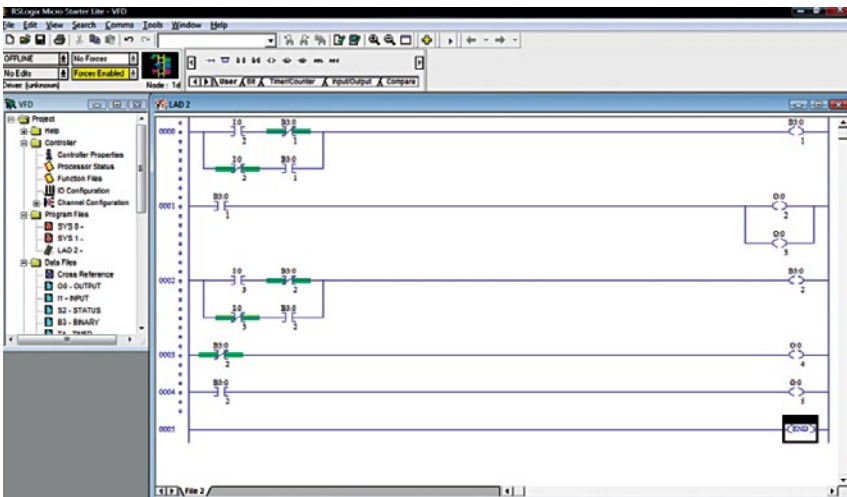


Fig. 9: Ladder program for control

Pr.01 set to 01 (external controls, via M0, M1)

Pr.38 set to 01 (M0, M1 set as run/stop and fwd/rev)

Set Pr.00 to 00 for controlling master frequency using the digital keypad, and 01 for controlling via potentiometer attached as shown in first pin connection diagram.

Pr.38 should be set to 01 as shown in above diagrams.

Once all these parameters are set, follow 2-wire mode operational steps for running the motor.

Using the PLC

Programmable logic controllers (PLCs) support digital input/output very effectively. So a PLC can also be used to control the operation of a VFD, hence to finally control the connected 3-phase induction motor.

Allen Bradley MicroLogix 1000 PLC is connected to Delta VFD-M and is programmed using Ladder programming using RS Logix. (Refer to the article published in May 2015 for more details on this.)

We connected M0 and M1 to O2



Fig. 10: Control board for motor control operation



Fig. 11: Allen Bradley PLC connection



Fig. 12: Varying the speed of the induction motor via potentiometer



Fig. 13: Three-phase induction motor

and O3 (outputs) of PLC and control O2 and O3 using Ladder Logic. Fig. 4 shows a logic defined for mode 01, that is, Pr.38 = 01. O:0.0/2 is connected to M0.

When I:0.0/2 is set on, it puts the motor in run mode. Now, even if I:0.0/2 is switched off, O:0.0/2 remains on due to the logic defined. It can only be stopped by pressing I:0.0/2 again.

I:0.0/3 controls O:0.0/5, which, in turn, is connected to M1, which decides the direction of rotation of the motor.

O:0.0/3 is the LED that turns on when the motor is in run mode.

O:0.0/5 is the LED that turns on when the motor is running in forward direction and turns off in reverse rotation. **EFY**